SOUMIL GOYAL

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Award-winning Mechanical Engineering student and former military officer seeking part-time or full-time internships. Offering extensive experience in advanced CAD/CAM, robotics, rapid prototyping, programming, and operational leadership.

EDUCATION

Georgia Institute of Technology: Honors Program | *B.S. in Mechanical Engineering* | GPA: 4.0/4.0 August 2026 – May 2029

MB Lamar High School, Houston, TX | *IB Diploma, Distinguished Achievement* | GPA: 4.0/4.0 August 2020 – June 2024

EXPERIENCE

Singapore Armed Forces | *Lieutenant, Platoon Commander* July 2024 – May 2026

- Promoted early to Lieutenant (the highest achievable rank for national service), commanding a 34-soldier platoon.
- Planned and executed ship-to-shore logistics alongside US, Thai, and South Korean forces for Exercise Cobra Gold 2026.
- Directed planning and safety as Supervising, Conducting, and Safety Officer for large-scale (1000+ soldiers) training conducts, earning the Battalion's "Safety Core Value" award and the 2026 Army Transport Formation "I am Safe" award.

Autodesk | *Industry Speaker, Student Ambassador* February 2024 – Present

- Selected to speak at the Autodesk University 2023 international engineering conference as the sole student presenter; Gave a 1-hour presentation to 50+ professionals and industry leaders on advancing the future of engineering education.
- Selected to consult with product developers at the 2024 TinkerCAD Summit to optimize advanced student CAD workflows.
- Planned and executed a 4-day CAD academy with 20+ students teaching advanced Fusion CAD/CAM features, generative design, and Autodesk User & Associate-level certification training.

ION Houston | *CNC Specialist Intern* June 2023 – June 2024

- Restored a 3-axis CNC machine for the professional makerspace at the ION Prototyping Lab Houston.
- Performed CAM, 3D Printing, and assembly of robotics showcase projects for Houston Robotics Group.

SKILLS

CAD/CAM: Autodesk Fusion, SolidWorks, CATIA, Onshape, AutoCAD. Generative Design, Static/Dynamic Simulations, Surfacing.
Manufacturing/Assembly: 3D Printing, CNC, Laser Cutting, Soldering, Microcontrollers, Raspberry Pi
Programming: C++, Python, Java, HTML/CSS/JS, MATLAB, Bash, TensorFlow/Keras (CNNs), OpenCV + YOLO V8, OpenAI API
Languages: English (Native), Hindi (Native), Mandarin (Professional Working Proficiency)
Certifications: Autodesk Fusion Certified User, OSHA-10 General Industry Safety Certification

PROJECTS

Highwater Haven | *$1000 First Prize* in the *'Make It Heal Design Competition'* amongst 99 entries December 2025 – June 2026

- Designed a hurricane-resistant, off-grid community refuge as a rapid-recovery space in response to Houston's flooding.
- Balanced structural integrity with aesthetics by using generative design iterations and FEA simulations.
- Integrated an ADA-compliant accessible layout by utilizing surface modeling for the building's organic architecture.

Samara Scouts | *Student Prize* in the *2026 'Dream a Better World'* competition amongst 92 entries April 2026 – June 2026

- Engineered a mass-manufacturable, air-dropped sensor platform to deliver early warning systems to remote/rugged terrains.
- Used 3D printing and rapid prototyping to optimize the maple-seed inspired biomimetic wing design.

SureShelter | *$300 Design and Make Prize* in the *2025 'Make It Temporary' Design Contest* February 2025 – May 2025

- Designed and engineered a modular, affordable, rapidly deployable temporary shelter for crisis response.
- Automated the generation of step-by-step CAD animations by programming a custom Fusion API Python script.

Robust Adaptive Network (RAN) | *Best Voted Idea* and in the *2022 Ambition Accelerator* May 2022 – May 2024

- Secured developmental backing from the Mayor of Houston's Office, Taco Bell Foundation, and Ashoka Foundation.
- Prototyped drone swarms and rovers designed to restore emergency communications and supply lines after natural disaster.
- Designed using Fusion. Used CNC machining and 3D printing for structural and mechanical testing.
- Prototyped proof-of-concept electronic subsystems using Raspberry Pi, microcontrollers, and custom Python scripts.

Pedal Power | *$750 grant winner* from the *National Wildlife Federation* October 2021 – April 2024

- Solo project to design and install human-powered laptop charging stations for community use.
- Used Autodesk Fusion, CNC laser cutting, and circuitry to deploy the custom 60W Pedal-Power station.

FIRST Robotics Competition | 2x *District Engineering Inspiration Award*, 2x *Industrial Design Award* August 2021 – May 2024

- Captained Team 2587, unifying designs from 4 schools across Onshape, Fusion, and AutoCAD and code across Java and Python (Raspberry Pi, OpenCV) leading to *2x State Championship qualification*.
- High quality CNC machining, laser cutting, 3D printing, sheet metal fabrication, and electronics work earned *the 2022 General Motors Industrial Design Award* and 10 other regional and district awards.

VEX Robotics Competition | *World Championship Inspiration All-Star Award, National Quarterfinalist* August 2021 – May 2024

- Captained my team to 27 awards across 3 years, the most in school history by leading rapid development in Fusion and extensive use of laser cutting custom structures.
- Programmed *UIL Texas State Championship's highest scoring autonomous program* by implementing object detection, odometry, and multithreading in C++.